

STRIDE Battery Status Monitor Board User Manual

The STRIDE Battery Status Monitor Board is designed to help monitor the condition and operation of the rollator battery system in a simple and reliable way. The board continuously tracks important battery information such as voltage, current flow, estimated charge usage, and fault conditions to help ensure safe and dependable operation of the STRIDE system.

This board is intended to work alongside the STRIDE motor and control system and was designed around a 24 V battery platform capable of supporting high-current motor loads.

1. Purpose of the Board

The Battery Status Monitor Board performs the following major functions:

- Monitors battery voltage
- Measures battery current draw
- Estimates battery usage and remaining charge
- Provides visual system status indication
- Communicates battery information to the main controller
- Helps detect abnormal operating conditions

The board was developed using the Analog Devices / LTC2944 battery monitoring IC for accurate battery tracking and coulomb counting functionality.

2. System Overview

The board operates by measuring:

- The voltage of the battery pack
- The amount of current flowing into or out of the battery
- The total charge consumed over time

A precision current sensing resistor (shunt resistor) is used to safely measure battery current.

The board communicates with the STM32 control system through I2C communication and can provide real-time battery information during rollator operation.

Real-Time Battery Monitoring

The board continuously measures:

- Battery voltage
- Current consumption
- Estimated battery usage
- Power draw

This allows the system using it to make smarter movement and safety decisions.

3. LED Indicators

The board includes two high-visibility RED LEDs for quick system feedback.

ON LED (Bright Red)

This LED indicates that the board is receiving power and operating normally.

Behavior

- ON continuously - Board powered correctly

- OFF - No power or power connection issue

This LED should normally remain ON whenever the STRIDE system is active.

ALERT LED (Bright Red)

This LED is used to indicate warning or fault conditions.

Possible Reasons for ALERT LED Activation

- Battery undervoltage condition
- Battery voltage outside the set threshold range
- Charge level below the programmed low-charge threshold
- Charge level above the programmed high-charge threshold
- Battery monitor temperature alert
- Accumulated charge register overflow
- Battery current outside the set threshold rang

Behavior

- OFF - Normal operation
- ON or flashing - Attention required

If the ALERT LED activates, the system should be inspected before continued operation.

4. Ratings

Supply Ratings

	Min Voltage	Max voltage	Min current(mA)	Max current(A)
Power Pins	5	37	12	0.5

The values on this table are based on specific component manufacturer's absolute ratings.

Measurement ratings

The board is designed for a 10 W to 100 W shunt resistor

	Min Voltage	Max voltage	Min current(mA)	Max current(A)
Sense pins	5	60	12	50

5. Normal Operation

During standard operation:

1. The ON LED should illuminate immediately after power-up.
2. The ALERT LED should remain OFF.
3. Battery information is automatically sent to the control system.
4. The system continuously tracks battery usage during operation.

No user interaction is normally required.

6. Start-Up Procedure

Step 1: Connect power supply and Microcontroller

Connect the battery Monitor to power supply via molex mating connectors. C connect the communication lines via 2-pin JST XH mating connectors on the board.

Step 2: Apply Power

Turn on the STRIDE system power and notice:

- ON LED turns ON
- ALERT LED remains OF

Once powered:

- The board automatically begins monitoring
- Battery measurements are updated continuously
- Data becomes available to the main controller

7. Safety Notes

- Do not reverse battery polarity
- Do not Short circuit board connections
- Do not exceed intended operating voltage. While the chip can handle up to 60 v, it is recommended that you don't surpass the absolute maximum ratings for this board
- Do not operate with exposed conductive wiring. Ensure your shunt resistor is covered and corresponding wiring is insulated
- The system can experience high motor current draw during acceleration or heavy load conditions. Care should always be taken when connecting batteries, servicing wiring and handling power connections

8. Troubleshooting

ON LED OFF

Possible causes:

- No battery power
- Loose power connection
- Damaged wiring

Recommended action:

- Verify battery voltage
- Check wiring connections
- Inspect fuse and connectors

ALERT LED ON

Possible causes:

- Low battery condition
- Excessive current draw
- Sensor fault
- Communication issue

Recommended action:

- Stop operation safely
- Inspect battery condition
- Verify cable connections

- Restart system if necessary

INCONSISTENT READINGS

Possible causes:

- Loose shunt resistor connections
- Electrical noise
- Communication interruption

Recommended action:

- Inspect all connectors
- Ensure wiring is secure
- Restart the control system

9. Integration with STRIDE System

The Battery Status Monitor Board is part of the larger STRIDE intelligent rollator platform, which includes:

- Motor control systems
- Safety sensors
- Embedded controllers
- Autonomous support functions

The battery monitoring system helps improve:

- User safety
- Reliability
- Runtime awareness
- Fault detection

The STRIDE system was developed as a smart assistive mobility platform for improved independence and safety.

10. Technical Summary

Feature	Description
Battery System	24 V
Monitoring Type	Voltage + Current + Charge Tracking
Communication	I2C
Status Indicators	Bright Red ON LED, Bright Red ALERT LED
Current Measurement	Precision Shunt Resistor
Main Battery Monitor IC	LTC2944
Intended Application	STRIDE Smart Rollator

11. User Notes

- The board operates automatically after power-up.
- No calibration is required during normal use.
- Always monitor the ALERT LED during operation.
- Disconnect power before servicing the system